

Biopreservation

- Biopreservation refers to extended storage life and enhanced safety of foods using the natural antimicrobial compounds that are of plant, animal and microbial origin and have been used in human food for long time, without any adverse effect on human health
- Biopreservation reduces the amounts of chemical preservatives as well as the intensity of heat treatments, both of which can otherwise negatively affect the food quality

Biopreservation

Extended storage life and enhanced safety of foods using the natural microflora and/or their antibacterial products

Application in food:

- 1. use bacterial strains**
- 2. add purified substance**
- 3. add fermentation liquor or concentrate**

Multiple Antimicrobial Effects of Lactic Acid Bacteria

- **Acids** - Lactate, acetate, formate, propionate
- **Diacetyl**, acetaldehyde
- **Reuterin** = aldehyde
- **Bacteriocins**
- ✓ Antimicrobial peptides
- ✓ kill or inhibit growth of closely related bacteria
- ✓ bacteriocins of Gram-positive bacteria are seldom active against Gram-negative bacteria

Commercially Available Products

- Bactoferm F-Lc (pediocin Chr-Hansen)
- ALTA 2351 (pediocin-Quest)
- Sakacin
- ALCMix1 (plantaricin-Danisco)
- Carnocin
- Micocin (carnocyclin-Griffith Laboratories)
- Piscicolin
- Carnobacteriocin

Selection of Cultures as Biopreservatives

- **Ability to produce antimicrobials in meats**
- **Limited sensory changes**
- **Limited acid production**
- **Weak protease activity**
- **Limited gas production**
- **Absence of slime production**

Micro-organisms
Protective Cultures



Food

- ✓ final products
- ✓ pre-stages
- ✓ or raw material



- **Inhibition of food pathogenic micro-organisms**
- growth control
- prevention of toxin-formation
- reduction of pathogens

and/or



- **Shelf life extension** based on targeted inhibition of specific spoilage micro-organisms



Biopreservation with fermentation end products of LAB

- ✓ Producing organic acids
- ✓ Hydrogen peroxide
- ✓ Diacetyl
- ✓ Antifungal compounds such as fatty acids or phenyl lactic acid
- ✓ Bacteriocins

ORGANIC ACIDS

Lactic acid

Propionic acid

Acetic acid

LOW MOLECULAR WEIGHT COMPOUNDS

Reuterin

Diacetyl

H₂O₂

Fatty acids

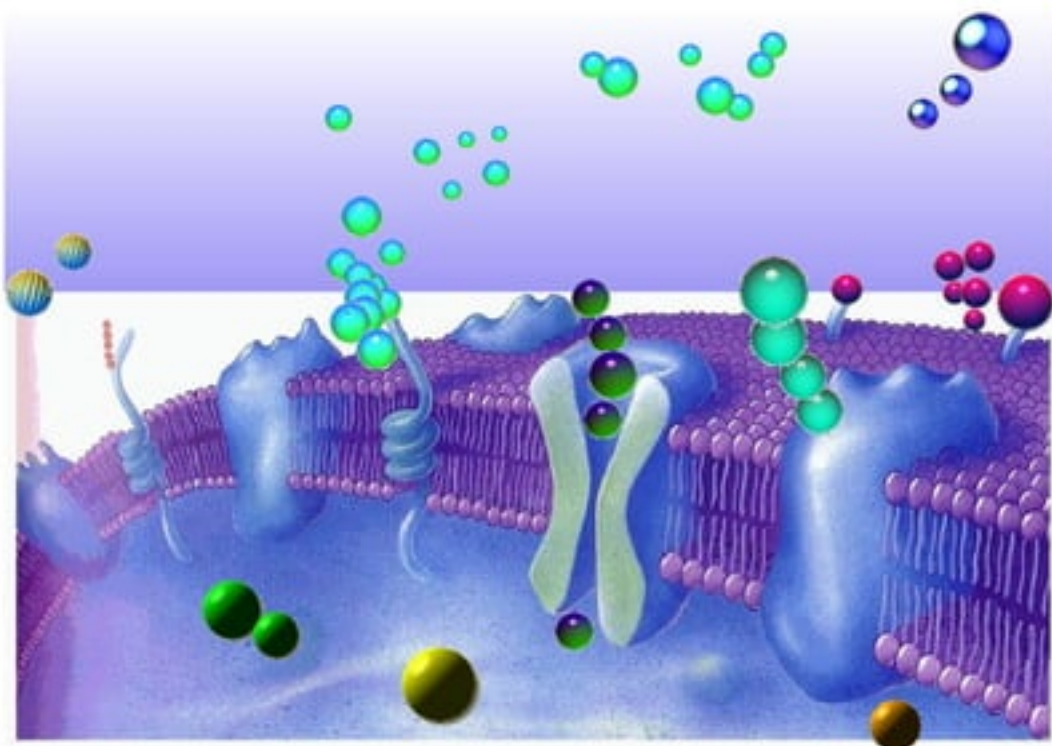
Phenyl lactic acids

Cyclic dipeptides

BACTERIOCINS (Antimicrobial Peptides)

What makes them anti-microbial?

Anti-microbial metabolites



- **Organic acids**
 - ✓ lactic acid
 - ✓ acetic acid
 - ✓ propionic acid...
- ➤ **Hydrogen peroxide**
- **Carbon dioxide**
- **Lacto peroxide**
- **Fatty acids**
- **Diacetyl**
- **Acetaldehyde**
- **Reuterin**
- **Other compounds of low molecular mass**
- **Bacteriocins**

- Weak acids Have More powerful antimicrobial activity at low pH than at neutral pH
- Acetic acid Is strongest inhibitor and has a wide range of inhibitory activity, inhibiting Yeast, Molds and bacteria
- Un-dissociated molecule is the toxic form of a weak acid

Undissociated form of organic acid diffuses across the cell membrane



Acid dissociates in the cytoplasm and releases the protons

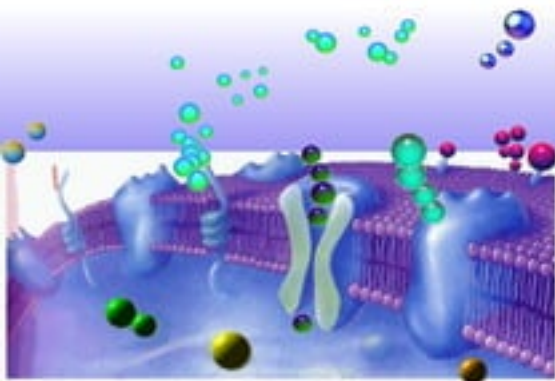


Leads to acidification and dissipation of pH gradient over the membrane causing the observed growth inhibition

Another Hypothesis

- ✓ Accumulation of anion is the cause of growth inhibition
- ✓ It reduces the rate of macromolecule synthesis and affects transport across the cell membrane

- more than anti-microbial metabolites!



**anti-microbial
metabolites**



**+ competitive
exclusion
effects**



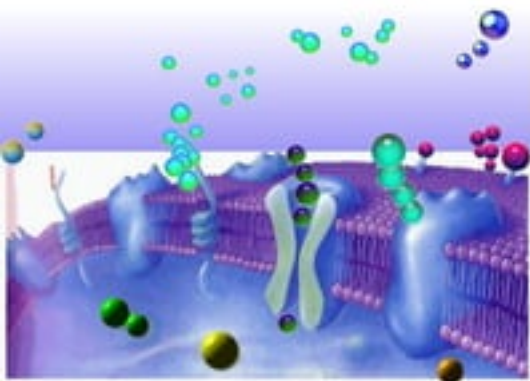
**Inhibition of
undesired
micro-organisms by
competing for
nutrients, oxygen, etc.
e.g. quorum sensing**

**+ further, so
far unknown
effects**



**= Inhibition of food pathogenic
and food spoilage micro-organisms**

A natural way



Bio preservation by protective cultures:

= Safer food



= Label friendly –
It's a 'culture'
and not a food preservative



= Positive image of
'biopreservation'-
'natural'



Protective Cultures - categories



Starter Cultures



Adjunct-Cultures



**Multi-functional
Cultures**



**Non-fermenting
Protective
Cultures**

**STARTER
CULTURE**

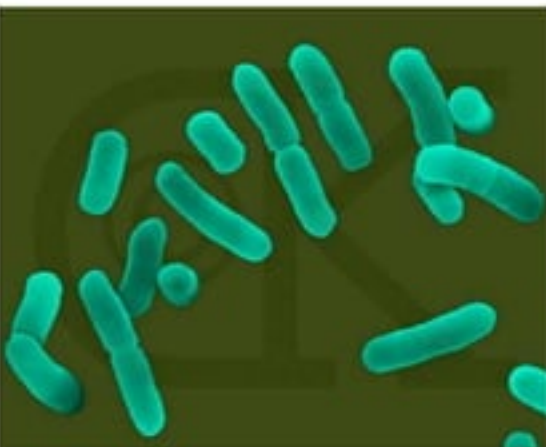
**PROTECTIVE
CULTURE**



**fermentation
processes:
Changes of
taste, flavour and
texture**

**no influence on
sensory
characteristics**

Protective cultures - are safe



Protective cultures are:

- ✓ generally recognised as safe (GRAS)
- ✓ there is no indication of a health risk of this bacterial group
- ✓ the use of lactic acid bacteria in biological preservation might even contribute to the health benefits of a product *)

• Lactic acid bacteria

Lactobacillus sp.

Lactococcus sp.

Pediococcus sp.

Carnobacterium sp.

Enterococcus sp.

Leuconostoc sp.

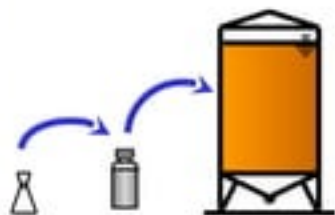
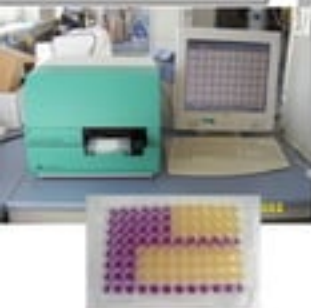
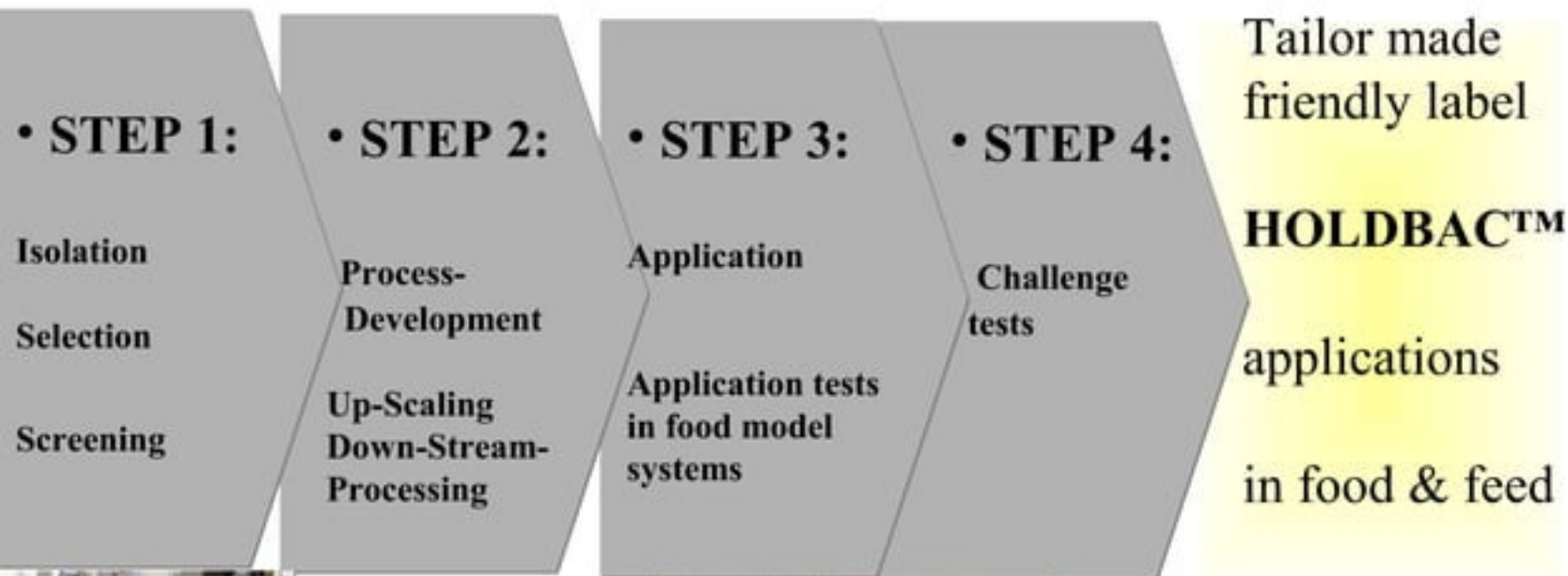
Weisella sp.

Vagococcus sp.

• *Propionibacteria*

*) Holzapfel et al., 1995, Int. J. Food Microbiol. 78: 119-131

Cultures - 4 steps from lab to customers

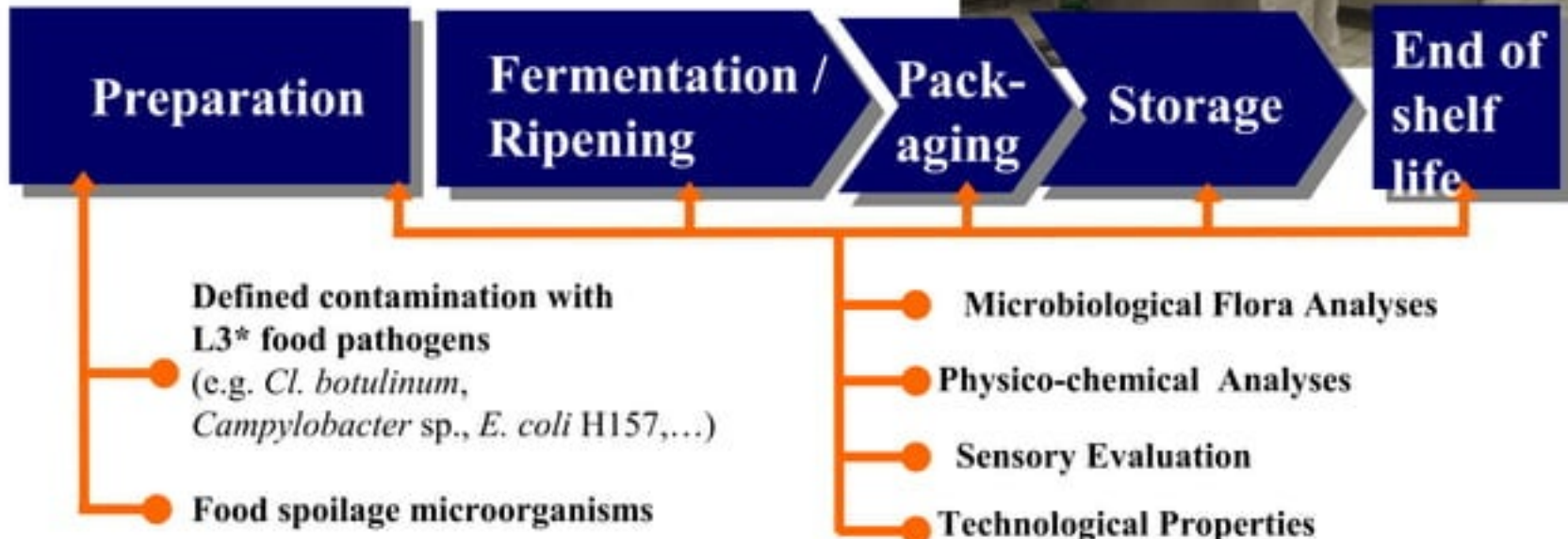


Food Model Systems

To reproduce the food products from
production to the end of shelf life:

Continuous Monitoring:

- Behaviour of indicator strains (growth – inhibition)
- Influence of the Food Protectants on technological and organoleptic properties in real food matrixes



Bacteriocins

- Bacteriocins are ribosomally synthesized single polypeptides or post-translationally modified ones that are usually inhibitory only to closely related bacterial species

Desirable properties that make Bacteriocins suitable for food preservation

- ✓ They are generally recognized as safe substances
- ✓ They are not active and nontoxic to eukaryotic cells
- ✓ They become inactivated by digestive proteases, having little influence on the gut microbiota
- ✓ They are usually pH and heat-tolerant
- ✓ They have a relatively broad antimicrobial spectrum against many food-borne pathogenic and spoilage bacteria
- ✓ They show a bactericidal mode of action, usually acting on the bacterial cytoplasmic membrane: no cross resistance with antibiotics
- ✓ Their genetic determinants are usually plasmid-encoded

Bacteriocins of LAB

Main category	Characteristics	Subcategory	Examples
Class I	Lantibiotics	Type A Type B	Nisin Mersacidin
Class II	Non-modified heat-stable bacteriocins containing peptides with molecular masses of 10 kDa	Subclass IIa Subclass IIb	Pediocin AcH Enterocin FH99 Pediocin 34 Lactacin F and Lactococcin G
Class III	Protein bacteriocins with molecular masses of 30 kDa		Helveticin J Lactacins A and B
Class IV	Bacteriocins that form large complexes with other macromolecules		Leuconosin S Lactococcin 27

Some useful bacteriocins for the dairy & food industry

Genus of Bacteria

Lactococcus lactis subsp. *lactis*

Pediococcus acidilactici

Pediococcus peantosaceus

Leuconostoc spp.

Lactobacillus sake

Lactobacillus plantarum

Lactobacillus helveticus

Carnobacterium piscicola



Bacteriocin Produced

Nisin A, Z

Pediocin PA-1, AcH

Pediocin 34

Leucocins

Sakacin A

Plantaricin

Helveticin J

Carnocin/piscicolin

**FACTORS PROMOTING
USE OF BACTERIOCINS
AS
BIOPRESERVATIVES**

Economical

**Safe and efficacious
use of nisin for > 40 years
in several countries**

**Consumer resistance
to traditional chemical
preservatives and
concern over
the safety of existing
food preservatives
such as sulfites
and nitrites**

**Advent of novel
bacteriocins with
broad spectrum of
activity from
food grade LAB**

**Effective in low
concentrations**

**Activity is not
lost in the
Presence of
food additives
and effective
in dairy Foods
during storage**

**Effective under
wide pH &
temperature range**

**Do not alter
acceptance
quality of
food and are
safe for human
consumption**

Bacteriocin Based Biopreservation Strategies

- Using a purified/ semi-purified bacteriocin preparation as an additive in food
- By incorporating an ingredient previously fermented with a bacteriocin-producing strain
- By using a bacteriocin-producing culture to replace all or part of a starter culture in fermented foods to produce the bacteriocin *in situ*.

Use of Purified/ Semi purified Bacteriocins

To date, the only commercially produced bacteriocins are:

- Nisin produced by *Lactococcus lactis* ssp. *lactis*
- **Pediocin PA-1, produced by *Pediococcus acidilactici***

The use of purified bacteriocins have to be labeled as additives and require regulatory approval.

Nisin is the only bacteriocin, approved for use as an antimicrobial in food by the Joint FAO/WHO Expert Committee on Food Additives

NISIN

Approved by FDA

Used in over 50 countries

Effective in a number of food systems

Wide range of inhibition spectrum

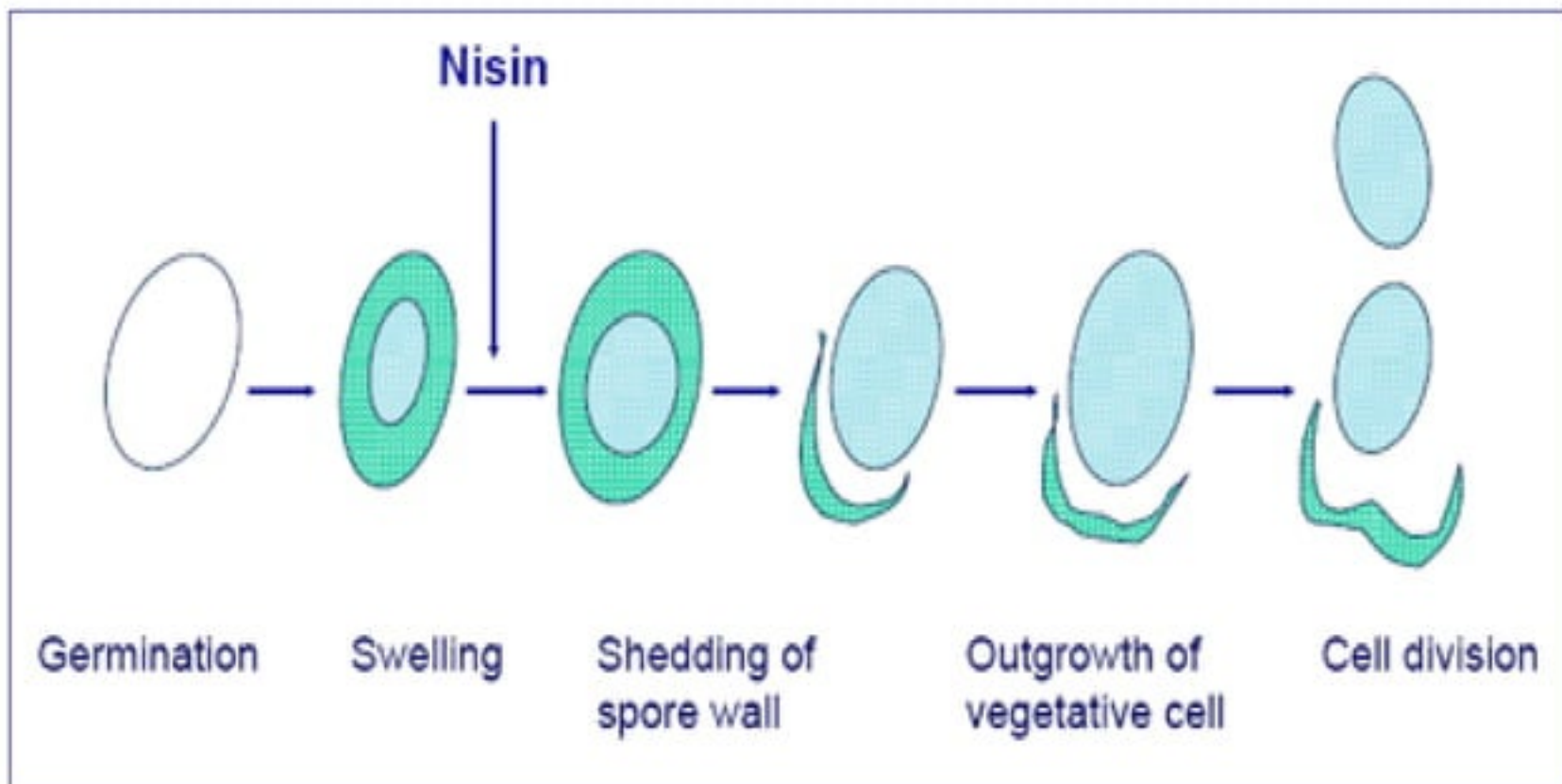
The level of nisin used depends on:

- ✓ Food composition
- ✓ Required shelf life
- ✓ Temperatures likely to be encountered during storage



Heat-resistant endospores

- ✓ Nisin affects spores after germination, preventing their outgrowth
- ✓ Does not normally kill bacterial spores



The realisation of Nisin as a food preservative

Nisin was used as a food preservative because:

- Nisin is non toxic
 - The producer strain *L. lactis* is regarded as safe (food-grade)
 - There is no apparent cross-resistance related to therapeutic antibiotics
 - It is degraded immediately during digestion
 - It is heat stable at low pH
-
- ✓ The **first commercial preparation of nisin (Nisaplin®)** was made in 1953 by **Aplin & Barrett Ltd. (now Danisco)**.
 - ✓ Originally intended only to control **clostridial spoilage in processed cheese**, the application was so successful it lead to worldwide interest.
 - ✓ In 1969 a joint **FAO-WHO** expert committee on food additives recognized nisin as a safe and legal food additive

Use of Nisin in the Biopreservation of Dairy Products

- Nisin is used in pasteurized, processed cheese products to prevent outgrowth of spores of *Clostridium tyrobutyricum*
- Used to extend the shelf life of dairy desserts which cannot be fully sterilized

Preservation of Khoa with Nisin (100 IU/gm)

Temperature	Shelf life
10 C	up to 90 days
22 C	up to 42 days
30 C	up to 28 days

Preservation of Kheer with Nisin (200 IU/gm)

Temperature	Shelf life (In days)		
	Control	Sterilized	Nisaplin added
37 C	2-3	3-4	8-10
4 C	10 -15	60-70	100-150

Nisin in Stirred Yoghurt

- **Addition of 50 IU nisin/g to yoghurt after preparation gave an acceptable product with increased shelf-life upto 10 days at refrigeration temperature**
- **Prevents subsequent over-acidification of the yogurt.**

Pediocins

POTENTIAL BACTERIOCINS AS BIOPRESERVATIVES

- ❖ Most pediocins are:
 - ✓ Thermostable proteins
 - ✓ Active over a wide range of pH
- ❖ Pediocin ACh:
 - ✓ Active against both spoilage and pathogenic organisms
 - ✓ *L. monocytogenes*, *Enterococcus faecalis*, *Staphylococcus aureus*, and *Clostridium perfringens*
- ❖ Pediocin PA-1:
 - ✓ Inhibits *Listeria* in dairy products such as cottage cheese, ice cream, and reconstituted dry milk

USE OF A BACTERIOCIN-PRODUCING CULTURE TO REPLACE ALL OR PART OF A STARTER CULTURE IN FERMENTED FOODS TO PRODUCE THE BACTERIOCIN *IN SITU*

- The use of cultures to produce bacteriocins *in situ* as a means of bio-preservation
- A more natural method of shelf-life extension and improving the safety

Bacteriocin Based Bioprotective Cultures

BS-10® Nisin producing *L. lactis* spp. *lactis*, Chris Hansen

BIOPROFIT™ *L. rhamnosus* LC705, BioGaia

BOVAMINE Meat Cultures™ Texas Tech University

HOLDBAC™ *L. plantarum*, *L. rhamnosus*, *L. sakei*, *L. paracasei* and
Propionibacterium freundenreichii spp. *Shermanii*,
DANISCO

MOVING TOWARDS BETTER TOMORROW.....

- Identify new bacteriocins for application in foods
- Altering the specificity of existing bacteriocins
- Increasing the level of bacteriocin production
- Development of bacteriocin producing lactic starters through gene transfer system
- Continued study of physical and chemical properties

Other Antimicrobial agents

- Antimicrobials are agents that kill microorganisms or inhibit their growth
- Biocontrol is the use of one or more organisms to inhibit or control other organism

In Food industry

- Microbial Antagonism
- Competitive Exclusion Technology
- Antimicrobial Proteins
- Antimicrobial Peptides
- Antimicrobials of plant origin
- Antimicrobials of animal origin
- Bacteriophages
- B'dellovibrio



S. Waksman

Microbial antagonism

- ▮ Microbial antagonism is the inhibition of undesired or pathogenic microorganisms caused by competition for nutrients, and by the production of antimicrobial metabolites.
- ▮ 'Microbial Interference' is also used which is a general nonspecific inhibition or destruction of microorganism by other members of the same habitat or environment.
- ▮ Lactic acid bacteria (LAB) have been used in food production as an effective method for extending safe storage of foodstuffs by simple fermentation
- ▮ Pure cultures of lactic acid bacteria (LAB): These organisms are particularly suitable as antagonistic microorganisms in food because of their ability to inhibit other foodborne bacteria by a variety of means, including production of organic acids, hydrogen peroxide and bacteriocins.
- ▮ Lactococcus, Streptococcus, Pediococcus, Leuconostoc, Lactobacillus and Carnobacterium are the genera most commonly used as starter cultures in the fermentation processes of milk, meat and vegetable products (Stiles & Hastings, 1991).

Competitive exclusion technology

- Competitive exclusion (CE) as a technology, involves the addition of a non-pathogenic bacterial culture to the intestinal tract of food animals in order to reduce colonization or decrease populations of pathogenic bacteria in the gastrointestinal tract (Callaway et al., 2004).
- The use of CE cultures has been studied extensively in poultry to reduce *Salmonella* and *Campylobacter* carriage (Chen & Stern, 2001; La Ragione & Woodward, 2003; Wagner, 2006; Zhang, Ma, & Doyle, 2007a, 2007b).

Antimicrobial proteins

- The production of one or more antimicrobial active metabolites is part of the complex mechanisms by which a culture becomes established in the presence of other competing organisms (Holzapfel et al., 1995).
- These metabolites can include clinical or therapeutic low-molecular weight antibiotics, lytic agents, toxins, bacteriolytic enzymes and other metabolic products such as hydrogen peroxide and diacetyl (Holzapfel et al., 1995; Kostrzynska & Bachand, 2006).

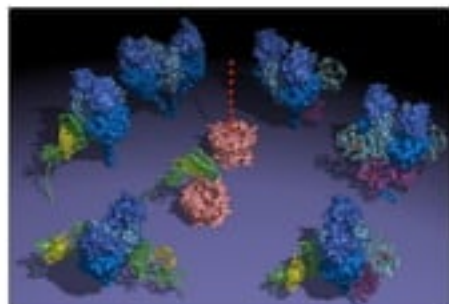
Bacteriocins

- These are proteins that exert an antimicrobial action against a range of microorganisms.
- Bacteriocins are ribosomally synthesized peptides whose responsible genes are frequently associated with transposons or on plasmids (Deegan, Cotter, Hill, & Ross, 2006).
- Their production can be related to the antagonism within a certain ecological niche, as the producer strain, being itself immune to its action, generally gains a competitive advantage.
- Bacteriocins have a wide antibacterial spectrum with feasible application in foods, such as meat and fish products, fruits and vegetables, cereals and beverages (Cleveland, Montville, Nes, & Chikindas, 2001).
- LAB-derived bacteriocins are generally recognized as safe (GRAS) and are attractive to the food industry because of their activity against key Gram-positive pathogens such as *Listeria monocytogenes* or *Staphylococcus aureus*.

- The most-studied bacteriocins in meat and meat products include nisin, enterocin AS-48, enterocins A and B, sakacin, leucocin A, and especially pediocin PA-1/AcH.
- These have shown promising results when used alone or in combination with several physicochemical treatments, modified atmosphere packaging, high hydrostatic pressure, (HHP), heat, and chemical preservatives, as an additional hurdle to control the proliferation of *L. monocytogenes* and other pathogens.

Antimicrobial peptides

- **Casecidin**, obtained from milk casein by chymosin digestion, exhibits activity against *Staphylococcus aureus*, *Sarcina*, *Bacillus subtilis*, *Diplococcus pneumoniae*, and *Streptococcus pyogenes* (Lahov and Regelson 1996).
- **Isracidin** is another casein-derived antimicrobial peptide. It was tested against infection with *Staphylococcus aureus* and *Candida albicans* in a mice (Lahov and Regelson 1996).
- **Lactoferricins** - several lactoferrin-derived antimicrobial peptides named **lactoferricin B** have been isolated (Hoek, Milne, Grieve, Dionysius, and Smith 1997; Shin et al. 1998).
- The hydrolyzate of lactoferrin exhibits broad antibacterial activity against both Gram-positive and Gram-negative bacteria.



- **OTAP-92** - Hen egg ovotransferrin-derived antimicrobial peptide that is active against *S. aureus* and *E. coli* has been isolated (Ibrahim, Sugimoto, and Aoki 2000).
- **Carnosine** (β -alanyl-L-histidine) and **Anserine** (N- β -alanyl-1-methyl-L-histidine) are endogenous antioxidative dipeptides found in skeletal muscle (Lynch and Kerry 2000). They are known to be the most abundant antioxidants in meats.
- **Microcins** are small peptides that show inhibition of Gram-negative bacteria.
- **Microcin J25 (MccJ25)** is active against *Salmonella spp.*, *Shigella spp.*, and *Escherichia coli* O157:H7.
- **MccJ25 variant** has been developed recently, which may be used as a food preservative against the Gram-negative pathogens (Pomares MF, Salomón RA, Pavlova O, Severinov K, Farías R, Vincent PA, 2009).

Limitations

- Not much research
- Cytotoxicity
- Allergenicity
- Long way for industrialization

Antimicrobials of Animal origin

Enzymes and other polypeptides with antimicrobial activity occur naturally in animals as well as plants.

Lytic enzymes-

❖ **Lysozyme** Its lysing activity has been demonstrated against *L. monocytogenes* and *C. botulinum* (Hughey and Johnson, 1987).

❖ **Glycosidases**

❖ **Amidase**

❖ **Lipases**

❖ **Nielsen (1991)** reported that lytic enzymes lysed *Campylobacter*, *E. coli*, *Salmonella*, and *V. parahaemolyticus*, indicating their potential use as natural food antimicrobials to reduce the risk of foodborne illness



Peroxidases and Oxidases

Lactoperoxidase

- ❖ **Lactoperoxidase** is the most abundant enzyme in bovine milk and is also produced in salivary glands of mammals.
- ❖ In raw milk, the lactoperoxidase system (LPS) inhibits lactic acid bacteria but is bactericidal to Gram-negative spoilage psychrotrophs and *Salmonella* (Reiter and Harnulv, 1984; Leyer and Johnson, 1993).
- ❖ The LPS is also active against Gram-positive pathogens such as *L. monocytogenes* (Siragusa and Johnson, 1989), *S. aureus* (Kamau et al., 1990), and *B. cereus* (Zajac et al., 1981).

Glucose oxidase

- ❖ Inhibition of growth of *Salmonella infantis*, *S. aureus*, *Clostridium perfringens*, and *B. cereus* by glucose oxidase was reported by Tiina and Sandholm (1989).



Antimicrobials of plant origin

- There are numerous natural compounds in plants that are known to prevent the growth of microorganisms
- Naturally occurring antimicrobial compounds are present in plant leaves, stems, barks, roots, flowers, and fruits



Allium

- Garlic (*Allium sativum*), onion (*A. cepa*), and leek (*A. porrum*) have allicin, a diallyl thiosulfinate (2-propenyl-2-propenethiol sulfinate), as an antimicrobial
- Work on antimicrobial activity of garlic and other *Allium* species has been done using foodborne pathogenic bacteria, mycotoxigenic molds, and spoilage microorganisms in general.
- Since sulfhydryl enzymes are common to all of these microorganisms, the spectrum of activity of *Allium* extracts is broad have effect against *Staphylococcus aureus* and *Bacillus species*, *Streptococcus*



Spices and herbs

- █ **Cinnamic aldehyde** (3-phenyl-2-propenal) has shown to be the major antimicrobial compound in **cinnamon**.
- █ Also inhibits mold growth (**Deans and Richie, 1987**),
- █ **Eugenol** [2-methoxy-4-(2-propenyl)phenol], a major constituent in **clove oil** and is present in considerable amounts in the essential oil of allspice, possesses antimicrobial activity.
- █ **Vanillin** (4-hydroxy-3-methoxybenzaldehyde), is a major constituent in **vanilla beans**, which is structurally similar to eugenol. Also **antimycotic**, Retard the growth of **yeasts** (**Boonchird and Flegel, 1982**) and **molds** (**Maruzzella and Liguori, 1958**). Vanillin is also **inhibitory** to ***L. monocytogenes*** (**Delaquis et al., 2005**)



Extracts and oils

- Recent research has focused on evaluating antimicrobial activities of extracts of aromatic plants not necessarily used as foods.
- Extracts of sumac (Nasar-Abbas and Halkman, 2004), gardenia and cedar wood (Friedman et al., 2002), and Lamiaceae species (Araujo et al., 2003), have been shown to exhibit inhibitory activities against foodborne pathogenic and spoilage microorganisms.
- Extracts of blueberries, crabapples, strawberries, red wines, grape juice, apple juice, and tea have been studied for their antiviral activity by Konowalchuk and Speirs (1978). These inactivated poliovirus, coxsackievirus, echovirus, reovirus and herpes simplex virus
- The primary inhibitors were thought to be tannins
- Tannic acid was antiviral against echovirus, poliovirus and herpes simplex virus

Phytoalexins

- Phytoalexins are low molecular weight compounds produced by higher plants in response to microbial infection and naturally occurring elicitors (Dixon et al., 1983).
- Phytoalexins are known to occur in leaves, fruits, seeds, roots, and tubers of a wide range of plants
- Glyceollin, coumestrol, and glycinol, are all phytoalexins produced by soybeans, which inhibit microbial membrane-associated processes (Weinstein and Albersheim, 1983; Amin et al., 1988).

MERITS AND DEMERITS OF BIO PRESERVATION

MERITS

Extended shelf-life of the meat and meat product

Decrease the of food borne pathogens transmission

Reduce use of chemical preservatives

Resistant organisms are controlled effectively
e.g. *L. monocytogenes*

Minimal handling and Natural way

Biopreservation



DEMERITS

Lack of research focus

Application deficit

Narrow activity of spectrum

Limited diffusion in solid matrix

Sometime difficult to apply

Commercialization

Consumers reluctance for fermented meat products

Regulatory approval